

Informativity of high-frequency bands on the place of articulation shift in retroflex sibilants produced by children

Supplementary Material

This supplement provides the complete results of the Linear Mixed-Effects (LME) models fitted separately for each of the 34 acoustic features (NE0–NE27, FNE12/23/34, and FNER ratios) and for each of the three analyzed sibilants. For every feature, a consolidated table reports fixed-effect estimates, standard errors, degrees of freedom, t- and p-values, and marginal/conditional R^2 for all three phonemes. Place of articulation (PoA) is coded relative to the Retroflex (normative) reference level; estimates for Dental, Interdental, and Postalveolar therefore represent the difference from the Retroflex intercept. Statistically significant fixed effects ($p < .05$) are highlighted in red.

Feature index:

- *NE0–NE27 = Noise Energy in consecutive 500 Hz subbands from 2–16 kHz;*
- *FNE12/23/34 = integrated Fricative Noise Energy between fricative formant frequency pairs;*
- *FNER(*i,j*) = ratio between the corresponding FNE measures.*

NE0

Noise Energy in the 2.0–2.5 kHz subband.

Table S1. Linear Mixed-Effects model results for NE0 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-323.202	4.103	85.860	-78.769	<0.001	0.051	0.323
Dental	-28.645	7.834	183.198	-3.656	<0.001	0.051	0.323
Interdental	13.959	13.511	178.032	1.033	0.303	0.051	0.323
Postalveolar	28.046	7.584	180.816	3.698	<0.001	0.051	0.323
Affricate /tʃ/							
(Intercept)	-306.482	5.343	5.187	-57.357	<0.001	0.027	0.239
Dental	-27.653	6.239	183.911	-4.432	<0.001	0.027	0.239
Interdental	-0.792	10.787	179.180	-0.073	0.942	0.027	0.239
Postalveolar	5.741	6.069	183.780	0.946	0.345	0.027	0.239
Fricative /z/							
(Intercept)	-364.617	6.354	25.036	-57.380	<0.001	0.049	0.417
Dental	-35.272	10.251	186.382	-3.441	0.001	0.049	0.417
Interdental	17.501	17.497	177.191	1.000	0.319	0.049	0.417
Postalveolar	28.827	9.808	179.169	2.939	0.004	0.049	0.417

NE1

Noise Energy in the 2.5–3.0 kHz subband.

Table S2. Linear Mixed-Effects model results for NE1 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-304.958	4.425	131.119	-68.924	<0.001	0.070	0.380
Dental	-55.414	9.017	183.743	-6.146	<0.001	0.070	0.380
Interdental	-18.323	15.574	179.329	-1.176	0.241	0.070	0.380
Postalveolar	8.513	8.735	181.593	0.975	0.331	0.070	0.380
Affricate /tʃ/							
(Intercept)	-314.240	4.937	6.670	-63.648	<0.001	0.045	0.260
Dental	-37.486	6.729	184.080	-5.571	<0.001	0.045	0.260
Interdental	-22.673	11.639	179.687	-1.948	0.053	0.045	0.260
Postalveolar	3.688	6.545	183.977	0.563	0.574	0.045	0.260
Fricative /z/							
(Intercept)	-353.286	6.580	35.574	-53.688	<0.001	0.059	0.410
Dental	-64.171	11.821	187.110	-5.429	<0.001	0.059	0.410
Interdental	-1.120	20.169	177.756	-0.056	0.956	0.059	0.410
Postalveolar	13.756	11.308	179.795	1.217	0.225	0.059	0.410

NE2

Noise Energy in the 3.0–3.5 kHz subband.

Table S3. Linear Mixed-Effects model results for NE2 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-284.982	7.735	25.556	-36.842	<0.001	0.058	0.461
Dental	-54.317	9.760	184.115	-5.565	<0.001	0.058	0.461
Interdental	-23.500	16.876	180.252	-1.392	0.165	0.058	0.461
Postalveolar	3.905	9.460	182.166	0.413	0.680	0.058	0.461
Affricate /tʃ/							
(Intercept)	-301.062	6.574	5.343	-45.794	<0.001	0.046	0.289
Dental	-38.568	7.701	183.506	-5.008	<0.001	0.046	0.289
Interdental	-39.088	13.329	179.649	-2.932	0.004	0.046	0.289
Postalveolar	2.331	7.490	183.439	0.311	0.756	0.046	0.289
Fricative /z/							
(Intercept)	-326.680	6.532	56.157	-50.008	<0.001	0.058	0.405
Dental	-71.780	12.710	187.794	-5.647	<0.001	0.058	0.405
Interdental	-13.922	21.686	178.387	-0.642	0.522	0.058	0.405
Postalveolar	4.677	12.158	180.444	0.385	0.701	0.058	0.405

NE3

Noise Energy in the 3.5–4.0 kHz subband.

Table S4. Linear Mixed-Effects model results for NE3 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-296.737	9.577	16.307	-30.984	<0.001	0.019	0.428
Dental	-28.427	7.822	184.042	-3.634	<0.001	0.019	0.428
Interdental	-25.176	13.492	178.934	-1.866	0.064	0.019	0.428
Postalveolar	-4.284	7.572	181.679	-0.566	0.572	0.019	0.428
Affricate /tʃ/							
(Intercept)	-310.579	7.970	4.075	-38.969	<0.001	0.021	0.253
Dental	-21.874	7.006	181.771	-3.122	0.002	0.021	0.253
Interdental	-31.715	12.114	177.161	-2.618	0.010	0.021	0.253
Postalveolar	2.986	6.814	181.645	0.438	0.662	0.021	0.253
Fricative /z/							
(Intercept)	-340.841	6.768	47.533	-50.361	<0.001	0.020	0.384
Dental	-41.160	12.465	187.222	-3.302	0.001	0.020	0.384
Interdental	-14.082	21.268	177.855	-0.662	0.509	0.020	0.384
Postalveolar	-3.209	11.924	179.899	-0.269	0.788	0.020	0.384

NE4

Noise Energy in the 4.0–4.5 kHz subband.

Table S5. Linear Mixed-Effects model results for NE4 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-321.764	9.039	16.814	-35.596	<0.001	0.005	0.394
Dental	-0.433	7.393	184.020	-0.059	0.953	0.005	0.394
Interdental	-21.168	12.742	178.543	-1.661	0.098	0.005	0.394
Postalveolar	-7.874	7.155	181.556	-1.100	0.273	0.005	0.394
Affricate /tʃ/							
(Intercept)	-331.425	8.787	3.991	-37.719	<0.001	0.006	0.276
Dental	-5.889	7.460	182.188	-0.789	0.431	0.006	0.276
Interdental	-23.319	12.911	178.273	-1.806	0.073	0.006	0.276
Postalveolar	-2.200	7.256	182.114	-0.303	0.762	0.006	0.276
Fricative /z/							
(Intercept)	-371.653	7.027	46.804	-52.888	<0.001	0.001	0.359
Dental	-6.600	12.908	187.418	-0.511	0.610	0.001	0.359
Interdental	3.539	22.007	177.763	0.161	0.872	0.001	0.359
Postalveolar	-7.670	12.340	179.926	-0.622	0.535	0.001	0.359

NE5

Noise Energy in the 4.5–5.0 kHz subband.

Table S6. Linear Mixed-Effects model results for NE5 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-340.210	9.080	16.186	-37.467	<0.001	0.016	0.395
Dental	18.888	7.128	183.067	2.650	0.009	0.016	0.395
Interdental	-4.612	12.280	177.379	-0.376	0.708	0.016	0.395
Postalveolar	-15.035	6.897	180.556	-2.180	0.031	0.016	0.395
Affricate /tʃ/							
(Intercept)	-354.298	7.749	4.525	-45.720	<0.001	0.018	0.286
Dental	23.302	7.776	183.885	2.997	0.003	0.018	0.286
Interdental	-9.925	13.461	180.193	-0.737	0.462	0.018	0.286
Postalveolar	-8.110	7.563	183.828	-1.072	0.285	0.018	0.286
Fricative /z/							
(Intercept)	-395.548	6.569	46.901	-60.212	<0.001	0.013	0.312
Dental	26.436	12.165	188.053	2.173	0.031	0.013	0.312
Interdental	21.568	20.672	177.137	1.043	0.298	0.013	0.312
Postalveolar	-12.150	11.601	179.863	-1.047	0.296	0.013	0.312

NE6

Noise Energy in the 5.0–5.5 kHz subband.

Table S7. Linear Mixed-Effects model results for NE6 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-343.288	10.623	14.737	-32.315	<0.001	0.017	0.407
Dental	19.161	7.274	180.937	2.634	0.009	0.017	0.407
Interdental	-10.397	12.521	174.914	-0.830	0.407	0.017	0.407
Postalveolar	-17.707	7.036	178.363	-2.517	0.013	0.017	0.407
Affricate /tʃ/							
(Intercept)	-366.228	9.232	4.080	-39.670	<0.001	0.021	0.287
Dental	28.348	8.116	184.288	3.493	0.001	0.021	0.287
Interdental	-11.751	14.047	180.389	-0.837	0.404	0.021	0.287
Postalveolar	-5.893	7.894	184.217	-0.747	0.456	0.021	0.287
Fricative /z/							
(Intercept)	-402.321	7.252	25.514	-55.474	<0.001	0.015	0.284
Dental	32.528	11.835	186.502	2.748	0.007	0.015	0.284
Interdental	21.253	20.060	174.686	1.060	0.291	0.015	0.284
Postalveolar	-11.086	11.265	177.872	-0.984	0.326	0.015	0.284

NE7

Noise Energy in the 5.5–6.0 kHz subband.

Table S8. Linear Mixed-Effects model results for NE7 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-353.150	10.779	15.777	-32.762	<0.001	0.013	0.410
Dental	19.624	8.339	181.845	2.353	0.020	0.013	0.410
Interdental	-10.541	14.371	176.387	-0.734	0.464	0.013	0.410
Postalveolar	-14.294	8.070	179.399	-1.771	0.078	0.013	0.410
Affricate /tʃ/							
(Intercept)	-376.060	8.774	4.662	-42.859	<0.001	0.019	0.310
Dental	26.760	9.072	184.257	2.950	0.004	0.019	0.310
Interdental	-5.211	15.714	180.992	-0.332	0.741	0.019	0.310
Postalveolar	-10.326	8.824	184.225	-1.170	0.243	0.019	0.310
Fricative /z/							
(Intercept)	-414.569	7.753	23.867	-53.471	<0.001	0.014	0.285
Dental	35.001	12.400	187.874	2.823	0.005	0.014	0.285
Interdental	14.252	21.016	175.975	0.678	0.499	0.014	0.285
Postalveolar	-11.045	11.802	179.182	-0.936	0.351	0.014	0.285

NE8

Noise Energy in the 6.0–6.5 kHz subband.

Table S9. Linear Mixed-Effects model results for NE8 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-387.067	9.004	22.765	-42.990	<0.001	0.012	0.446
Dental	16.329	10.457	182.902	1.562	0.120	0.012	0.446
Interdental	19.139	18.081	179.040	1.059	0.291	0.012	0.446
Postalveolar	-13.808	10.135	180.956	-1.362	0.175	0.012	0.446
Affricate /tʃ/							
(Intercept)	-400.560	8.149	6.035	-49.157	<0.001	0.017	0.379
Dental	23.243	10.304	183.718	2.256	0.025	0.017	0.379
Interdental	14.921	17.873	181.460	0.835	0.405	0.017	0.379
Postalveolar	-11.982	10.022	183.729	-1.196	0.233	0.017	0.379
Fricative /z/							
(Intercept)	-454.223	8.113	34.470	-55.986	<0.001	0.018	0.349
Dental	31.201	14.240	187.268	2.191	0.030	0.018	0.349
Interdental	46.273	24.248	177.136	1.908	0.058	0.018	0.349
Postalveolar	-9.568	13.601	179.509	-0.703	0.483	0.018	0.349

NE9

Noise Energy in the 6.5–7.0 kHz subband.

Table S10. Linear Mixed-Effects model results for NE9 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-402.518	8.708	24.717	-46.226	<0.001	0.016	0.437
Dental	14.642	10.630	182.710	1.377	0.170	0.016	0.437
Interdental	23.626	18.379	178.811	1.285	0.200	0.016	0.437
Postalveolar	-19.297	10.303	180.750	-1.873	0.063	0.016	0.437
Affricate /tʃ/							
(Intercept)	-414.558	8.171	5.749	-50.734	<0.001	0.021	0.371
Dental	18.827	10.043	184.000	1.875	0.062	0.021	0.371
Interdental	24.997	17.417	181.615	1.435	0.153	0.021	0.371
Postalveolar	-18.200	9.768	184.006	-1.863	0.064	0.021	0.371
Fricative /z/							
(Intercept)	-471.565	7.509	45.699	-62.801	<0.001	0.025	0.337
Dental	26.236	13.845	188.215	1.895	0.060	0.025	0.337
Interdental	65.438	23.553	177.699	2.778	0.006	0.025	0.337
Postalveolar	-12.186	13.214	180.236	-0.922	0.358	0.025	0.337

NE10

Noise Energy in the 7.0–7.5 kHz subband.

Table S11. Linear Mixed-Effects model results for NE10 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-411.856	7.972	23.196	-51.661	<0.001	0.030	0.392
Dental	30.665	9.223	181.387	3.325	0.001	0.030	0.392
Interdental	16.238	15.922	176.746	1.020	0.309	0.030	0.392
Postalveolar	-18.776	8.933	179.175	-2.102	0.037	0.030	0.392
Affricate /tʃ/							
(Intercept)	-419.584	6.894	5.719	-60.863	<0.001	0.018	0.287
Dental	18.217	8.453	184.061	2.155	0.032	0.018	0.287
Interdental	14.810	14.639	180.566	1.012	0.313	0.018	0.287
Postalveolar	-16.800	8.222	184.017	-2.043	0.042	0.018	0.287
Fricative /z/							
(Intercept)	-480.770	7.059	40.272	-68.107	<0.001	0.028	0.310
Dental	42.083	12.800	187.811	3.288	0.001	0.028	0.310
Interdental	53.887	21.730	176.544	2.480	0.014	0.028	0.310
Postalveolar	-8.822	12.198	179.442	-0.723	0.470	0.028	0.310

NE11

Noise Energy in the 7.5–8.0 kHz subband.

Table S12. Linear Mixed-Effects model results for NE11 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-416.950	9.872	21.878	-42.235	<0.001	0.038	0.458
Dental	40.755	10.877	182.205	3.747	<0.001	0.038	0.458
Interdental	6.701	18.803	178.223	0.356	0.722	0.038	0.458
Postalveolar	-23.407	10.541	180.216	-2.221	0.028	0.038	0.458
Affricate /tʃ/							
(Intercept)	-433.781	8.144	6.503	-53.266	<0.001	0.032	0.361
Dental	30.714	10.723	183.012	2.864	0.005	0.032	0.361
Interdental	27.691	18.593	180.490	1.489	0.138	0.032	0.361
Postalveolar	-21.566	10.430	183.013	-2.068	0.040	0.032	0.361
Fricative /z/							
(Intercept)	-486.836	8.414	38.692	-57.858	<0.001	0.032	0.374
Dental	56.232	14.978	185.757	3.754	<0.001	0.032	0.374
Interdental	39.558	25.529	176.055	1.550	0.123	0.032	0.374
Postalveolar	-11.463	14.316	178.256	-0.801	0.424	0.032	0.374

NE12

Noise Energy in the 8.0–8.5 kHz subband.

Table S13. Linear Mixed-Effects model results for NE12 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-438.171	9.297	33.714	-47.130	<0.001	0.041	0.535
Dental	42.029	13.383	182.472	3.140	0.002	0.041	0.535
Interdental	8.267	23.186	179.694	0.357	0.722	0.041	0.535
Postalveolar	-28.120	12.985	180.978	-2.166	0.032	0.041	0.535
Affricate /tʃ/							
(Intercept)	-452.436	6.985	13.550	-64.770	<0.001	0.039	0.417
Dental	28.254	11.833	183.065	2.388	0.018	0.039	0.417
Interdental	31.796	20.533	181.135	1.549	0.123	0.039	0.417
Postalveolar	-28.733	11.509	183.085	-2.497	0.013	0.039	0.417
Fricative /z/							
(Intercept)	-510.489	8.564	64.888	-59.608	<0.001	0.033	0.436
Dental	58.979	16.554	185.704	3.563	<0.001	0.033	0.436
Interdental	34.337	28.313	177.299	1.213	0.227	0.033	0.436
Postalveolar	-13.547	15.865	178.975	-0.854	0.394	0.033	0.436

NE13

Noise Energy in the 8.5–9.0 kHz subband.

Table S14. Linear Mixed-Effects model results for NE13 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-464.627	8.196	69.603	-56.688	<0.001	0.049	0.589
Dental	42.476	14.723	182.482	2.885	0.004	0.049	0.589
Interdental	29.838	25.532	180.316	1.169	0.244	0.049	0.589
Postalveolar	-31.563	14.292	181.277	-2.208	0.028	0.049	0.589
Affricate /tʃ/							
(Intercept)	-471.385	6.733	20.759	-70.011	<0.001	0.051	0.476
Dental	29.689	12.373	182.807	2.400	0.017	0.051	0.476
Interdental	42.590	21.481	181.269	1.983	0.049	0.051	0.476
Postalveolar	-32.481	12.034	182.833	-2.699	0.008	0.051	0.476
Fricative /z/							
(Intercept)	-537.363	9.047	60.276	-59.397	<0.001	0.037	0.480
Dental	54.136	17.286	185.723	3.132	0.002	0.037	0.480
Interdental	58.530	29.619	178.015	1.976	0.050	0.037	0.480
Postalveolar	-15.519	16.591	179.439	-0.935	0.351	0.037	0.480

NE14

Noise Energy in the 9.0–9.5 kHz subband.

Table S15. Linear Mixed-Effects model results for NE14 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-471.028	8.723	69.985	-53.997	<0.001	0.056	0.610
Dental	50.597	15.696	182.574	3.224	0.001	0.056	0.610
Interdental	40.478	27.226	180.542	1.487	0.139	0.056	0.610
Postalveolar	-30.916	15.239	181.436	-2.029	0.044	0.056	0.610
Affricate /tʃ/							
(Intercept)	-478.600	6.794	24.936	-70.442	<0.001	0.061	0.487
Dental	38.493	12.823	182.827	3.002	0.003	0.061	0.487
Interdental	50.229	22.265	181.335	2.256	0.025	0.061	0.487
Postalveolar	-30.958	12.473	182.853	-2.482	0.014	0.061	0.487
Fricative /z/							
(Intercept)	-546.860	8.945	84.359	-61.139	<0.001	0.047	0.501
Dental	63.333	17.934	185.817	3.531	0.001	0.047	0.501
Interdental	71.660	30.753	178.394	2.330	0.021	0.047	0.501
Postalveolar	-13.239	17.224	179.722	-0.769	0.443	0.047	0.501

NE15

Noise Energy in the 9.5–10.0 kHz subband.

Table S16. Linear Mixed-Effects model results for NE15 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-476.255	8.321	62.361	-57.233	<0.001	0.066	0.573
Dental	61.649	14.498	182.807	4.252	<0.001	0.066	0.573
Interdental	30.874	25.134	180.429	1.228	0.221	0.066	0.573
Postalveolar	-24.945	14.071	181.498	-1.773	0.078	0.066	0.573
Affricate /tʃ/							
(Intercept)	-484.517	6.693	23.231	-72.389	<0.001	0.067	0.469
Dental	46.669	12.494	182.965	3.735	<0.001	0.067	0.469
Interdental	49.007	21.689	181.320	2.260	0.025	0.067	0.469
Postalveolar	-28.159	12.152	182.990	-2.317	0.022	0.067	0.469
Fricative /z/							
(Intercept)	-551.381	8.412	83.857	-65.546	<0.001	0.060	0.487
Dental	77.283	16.870	185.968	4.581	<0.001	0.060	0.487
Interdental	67.158	28.902	178.194	2.324	0.021	0.060	0.487
Postalveolar	-8.216	16.189	179.640	-0.507	0.612	0.060	0.487

NE16

Noise Energy in the 10.0–10.5 kHz subband.

Table S17. Linear Mixed-Effects model results for NE16 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-486.666	6.671	66.091	-72.955	<0.001	0.084	0.498
Dental	68.159	11.854	182.937	5.750	<0.001	0.084	0.498
Interdental	11.680	20.522	179.777	0.569	0.570	0.084	0.498
Postalveolar	-18.736	11.497	181.272	-1.630	0.105	0.084	0.498
Affricate /tʃ/							
(Intercept)	-495.018	5.961	21.581	-83.048	<0.001	0.058	0.429
Dental	46.178	11.033	182.940	4.185	<0.001	0.058	0.429
Interdental	28.982	19.146	181.025	1.514	0.132	0.058	0.429
Postalveolar	-20.955	10.731	182.961	-1.953	0.052	0.058	0.429
Fricative /z/							
(Intercept)	-564.054	7.361	66.963	-76.626	<0.001	0.069	0.432
Dental	82.784	14.316	185.080	5.783	<0.001	0.069	0.432
Interdental	52.173	24.453	176.215	2.134	0.034	0.069	0.432
Postalveolar	-1.835	13.706	178.075	-0.134	0.894	0.069	0.432

NE17

Noise Energy in the 10.5–11.0 kHz subband.

Table S18. Linear Mixed-Effects model results for NE17 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-496.445	6.721	86.585	-73.867	<0.001	0.074	0.520
Dental	63.250	12.677	182.832	4.989	<0.001	0.074	0.520
Interdental	25.276	21.960	179.995	1.151	0.251	0.074	0.520
Postalveolar	-20.075	12.299	181.309	-1.632	0.104	0.074	0.520
Affricate /tʃ/							
(Intercept)	-502.740	5.628	47.749	-89.333	<0.001	0.058	0.454
Dental	42.578	11.472	182.772	3.711	<0.001	0.058	0.454
Interdental	36.850	19.914	181.080	1.850	0.066	0.058	0.454
Postalveolar	-22.803	11.159	182.798	-2.044	0.042	0.058	0.454
Fricative /z/							
(Intercept)	-571.766	7.887	44.919	-72.499	<0.001	0.068	0.445
Dental	75.916	14.317	185.458	5.303	<0.001	0.068	0.445
Interdental	66.380	24.467	176.779	2.713	0.007	0.068	0.445
Postalveolar	-3.791	13.712	178.561	-0.276	0.783	0.068	0.445

NE18

Noise Energy in the 11.0–11.5 kHz subband.

Table S19. Linear Mixed-Effects model results for NE18 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-499.109	7.163	128.350	-69.682	<0.001	0.067	0.576
Dental	60.125	14.458	182.632	4.159	<0.001	0.067	0.576
Interdental	38.639	25.068	180.364	1.541	0.125	0.067	0.576
Postalveolar	-22.209	14.034	181.376	-1.583	0.115	0.067	0.576
Affricate /tʃ/							
(Intercept)	-502.528	5.716	123.939	-87.911	<0.001	0.053	0.486
Dental	40.275	12.518	182.656	3.217	0.002	0.053	0.486
Interdental	40.993	21.736	181.203	1.886	0.061	0.053	0.486
Postalveolar	-24.249	12.176	182.689	-1.992	0.048	0.053	0.486
Fricative /z/							
(Intercept)	-570.687	8.556	60.917	-66.701	<0.001	0.064	0.512
Dental	74.342	16.349	184.895	4.547	<0.001	0.064	0.512
Interdental	76.373	28.035	177.509	2.724	0.007	0.064	0.512
Postalveolar	-2.623	15.701	178.829	-0.167	0.867	0.064	0.512

NE19

Noise Energy in the 11.5–12.0 kHz subband.

Table S20. Linear Mixed-Effects model results for NE19 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-499.272	7.401	106.243	-67.463	<0.001	0.060	0.570
Dental	60.524	14.426	182.518	4.196	<0.001	0.060	0.570
Interdental	27.607	25.011	180.207	1.104	0.271	0.060	0.570
Postalveolar	-18.754	14.002	181.242	-1.339	0.182	0.060	0.570
Affricate /tʃ/							
(Intercept)	-502.516	6.140	54.955	-81.846	<0.001	0.054	0.485
Dental	45.305	12.606	182.774	3.594	<0.001	0.054	0.485
Interdental	38.506	21.888	181.304	1.759	0.080	0.054	0.485
Postalveolar	-20.837	12.261	182.803	-1.699	0.091	0.054	0.485
Fricative /z/							
(Intercept)	-568.988	8.504	81.216	-66.907	<0.001	0.064	0.541
Dental	78.748	16.944	184.595	4.648	<0.001	0.064	0.541
Interdental	72.237	29.091	177.707	2.483	0.014	0.064	0.541
Postalveolar	4.911	16.289	178.871	0.302	0.763	0.064	0.541

NE20

Noise Energy in the 12.0–12.5 kHz subband.

Table S21. Linear Mixed-Effects model results for NE20 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-504.174	6.761	119.641	-74.569	<0.001	0.067	0.537
Dental	63.340	13.515	182.497	4.687	<0.001	0.067	0.537
Interdental	21.079	23.421	179.899	0.900	0.369	0.067	0.537
Postalveolar	-17.871	13.115	181.084	-1.363	0.175	0.067	0.537
Affricate /tʃ/							
(Intercept)	-505.857	5.804	60.880	-87.161	<0.001	0.057	0.464
Dental	46.385	12.034	182.779	3.854	<0.001	0.057	0.464
Interdental	36.295	20.892	181.165	1.737	0.084	0.057	0.464
Postalveolar	-20.766	11.705	182.808	-1.774	0.078	0.057	0.464
Fricative /z/							
(Intercept)	-573.156	8.200	65.638	-69.894	<0.001	0.070	0.518
Dental	82.248	15.885	184.587	5.178	<0.001	0.070	0.518
Interdental	67.439	27.242	177.259	2.476	0.014	0.070	0.518
Postalveolar	7.005	15.257	178.562	0.459	0.647	0.070	0.518

NE21

Noise Energy in the 12.5–13.0 kHz subband.

Table S22. Linear Mixed-Effects model results for NE21 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-514.602	6.628	118.604	-77.638	<0.001	0.062	0.531
Dental	57.582	13.256	182.418	4.344	<0.001	0.062	0.531
Interdental	23.203	22.971	179.787	1.010	0.314	0.062	0.531
Postalveolar	-20.484	12.863	180.990	-1.592	0.113	0.062	0.531
Affricate /tʃ/							
(Intercept)	-515.356	5.474	123.396	-94.141	<0.001	0.057	0.466
Dental	44.635	12.049	182.578	3.704	<0.001	0.057	0.466
Interdental	41.671	20.919	180.990	1.992	0.048	0.057	0.466
Postalveolar	-20.829	11.720	182.612	-1.777	0.077	0.057	0.466
Fricative /z/							
(Intercept)	-583.689	8.132	50.714	-71.778	<0.001	0.071	0.504
Dental	81.452	15.064	185.304	5.407	<0.001	0.071	0.504
Interdental	59.208	25.817	177.675	2.293	0.023	0.071	0.504
Postalveolar	2.458	14.460	179.074	0.170	0.865	0.071	0.504

NE22

Noise Energy in the 13.0–13.5 kHz subband.

Table S23. Linear Mixed-Effects model results for NE22 (/ʃ/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /ʃ/							
(Intercept)	-517.277	6.822	130.126	-75.821	<0.001	0.046	0.544
Dental	46.301	13.802	182.418	3.355	0.001	0.046	0.544
Interdental	27.788	23.925	179.977	1.161	0.247	0.046	0.544
Postalveolar	-21.521	13.395	181.079	-1.607	0.110	0.046	0.544
Affricate /tʃ/							
(Intercept)	-515.977	5.605	95.627	-92.048	<0.001	0.040	0.451
Dental	32.208	12.115	182.615	2.658	0.009	0.040	0.451
Interdental	37.107	21.032	180.984	1.764	0.079	0.040	0.451
Postalveolar	-22.424	11.784	182.646	-1.903	0.059	0.040	0.451
Fricative /z/							
(Intercept)	-583.803	8.433	45.628	-69.225	<0.001	0.059	0.502
Dental	69.521	15.310	185.054	4.541	<0.001	0.059	0.502
Interdental	64.676	26.242	177.490	2.465	0.015	0.059	0.502
Postalveolar	-1.633	14.698	178.868	-0.111	0.912	0.059	0.502

NE23

Noise Energy in the 13.5–14.0 kHz subband.

Table S24. Linear Mixed-Effects model results for NE23 (/ʃ/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /ʃ/							
(Intercept)	-522.434	7.357	126.620	-71.008	<0.001	0.038	0.575
Dental	43.512	14.840	182.503	2.932	0.004	0.038	0.575
Interdental	26.273	25.736	180.329	1.021	0.309	0.038	0.575
Postalveolar	-20.569	14.406	181.294	-1.428	0.155	0.038	0.575
Affricate /tʃ/							
(Intercept)	-520.229	5.586	170.343	-93.125	<0.001	0.037	0.468
Dental	32.597	12.683	182.352	2.570	0.011	0.037	0.468
Interdental	30.609	22.022	180.861	1.390	0.166	0.037	0.468
Postalveolar	-23.452	12.337	182.400	-1.901	0.059	0.037	0.468
Fricative /z/							
(Intercept)	-589.125	9.212	39.499	-63.953	<0.001	0.053	0.528
Dental	65.775	16.308	184.678	4.033	<0.001	0.053	0.528
Interdental	69.233	27.983	177.571	2.474	0.014	0.053	0.528
Postalveolar	-0.557	15.670	178.801	-0.036	0.972	0.053	0.528

NE24

Noise Energy in the 14.0–14.5 kHz subband.

Table S25. Linear Mixed-Effects model results for NE24 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-524.955	7.578	134.709	-69.270	<0.001	0.036	0.584
Dental	44.722	15.385	182.310	2.907	0.004	0.036	0.584
Interdental	16.571	26.684	180.218	0.621	0.535	0.036	0.584
Postalveolar	-20.174	14.936	181.142	-1.351	0.178	0.036	0.584
Affricate /tʃ/							
(Intercept)	-523.403	5.313	176.129	-98.505	<0.001	0.036	0.438
Dental	33.375	12.166	182.705	2.743	0.007	0.036	0.438
Interdental	27.866	21.118	181.031	1.320	0.189	0.036	0.438
Postalveolar	-22.131	11.834	182.765	-1.870	0.063	0.036	0.438
Fricative /z/							
(Intercept)	-591.124	9.640	36.995	-61.320	<0.001	0.049	0.542
Dental	68.489	16.806	184.817	4.075	<0.001	0.049	0.542
Interdental	55.348	28.856	177.952	1.918	0.057	0.049	0.542
Postalveolar	-2.435	16.157	179.106	-0.151	0.880	0.049	0.542

NE25

Noise Energy in the 14.5–15.0 kHz subband.

Table S26. Linear Mixed-Effects model results for NE25 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-529.114	7.810	127.841	-67.753	<0.001	0.036	0.589
Dental	46.158	15.707	182.282	2.939	0.004	0.036	0.589
Interdental	10.266	27.244	180.223	0.377	0.707	0.036	0.589
Postalveolar	-20.305	15.249	181.130	-1.332	0.185	0.036	0.589
Affricate /tʃ/							
(Intercept)	-526.431	5.336	183.635	-98.663	<0.001	0.063	—
Dental	34.630	12.253	182.819	2.826	0.005	0.063	—
Interdental	24.630	21.271	181.161	1.158	0.248	0.063	—
Postalveolar	-23.336	11.919	182.896	-1.958	0.052	0.063	—
Fricative /z/							
(Intercept)	-593.981	9.847	38.618	-60.319	<0.001	0.044	0.558
Dental	67.624	17.344	183.936	3.899	<0.001	0.044	0.558
Interdental	50.295	29.802	177.380	1.688	0.093	0.044	0.558
Postalveolar	-0.618	16.685	178.447	-0.037	0.971	0.044	0.558

NE26

Noise Energy in the 15.0–15.5 kHz subband.

Table S27. Linear Mixed-Effects model results for NE26 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-530.716	7.731	145.862	-68.645	<0.001	0.032	0.586
Dental	43.405	15.927	182.323	2.725	0.007	0.032	0.586
Interdental	11.458	27.625	180.268	0.415	0.679	0.032	0.586
Postalveolar	-20.301	15.462	181.173	-1.313	0.191	0.032	0.586
Affricate /tʃ/							
(Intercept)	-528.546	5.498	176.482	-96.129	<0.001	0.031	0.442
Dental	31.329	12.545	182.847	2.497	0.013	0.031	0.442
Interdental	31.168	21.778	181.215	1.431	0.154	0.031	0.442
Postalveolar	-19.837	12.203	182.910	-1.626	0.106	0.031	0.442
Fricative /z/							
(Intercept)	-594.081	9.721	48.112	-61.114	<0.001	0.041	0.568
Dental	65.144	17.865	184.923	3.646	<0.001	0.041	0.568
Interdental	54.102	30.713	178.540	1.762	0.080	0.041	0.568
Postalveolar	1.327	17.193	179.554	0.077	0.939	0.041	0.568

NE27

Noise Energy in the 15.5–16.0 kHz subband.

Table S28. Linear Mixed-Effects model results for NE27 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-528.649	7.632	165.934	-69.267	<0.001	0.027	0.589
Dental	36.967	16.146	182.262	2.289	0.023	0.027	0.589
Interdental	21.229	28.008	180.258	0.758	0.449	0.027	0.589
Postalveolar	-20.345	15.677	181.138	-1.298	0.196	0.027	0.589
Affricate /tʃ/							
(Intercept)	-525.847	5.450	174.133	-96.487	<0.001	0.033	0.434
Dental	28.682	12.387	182.979	2.315	0.022	0.033	0.434
Interdental	35.250	21.502	181.284	1.639	0.103	0.033	0.434
Postalveolar	-22.241	12.049	183.034	-1.846	0.067	0.033	0.434
Fricative /z/							
(Intercept)	-590.814	9.282	71.084	-63.649	<0.001	0.037	0.572
Dental	57.360	18.166	184.629	3.158	0.002	0.037	0.572
Interdental	61.058	31.239	178.379	1.955	0.052	0.037	0.572
Postalveolar	1.219	17.487	179.357	0.070	0.944	0.037	0.572

FNE12

Integrated Fricative Noise Energy between FF1 (2 kHz) and FF2 (4 kHz).

Table S29. Linear Mixed-Effects model results for FNE12 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-886.244	19.277	22.707	-45.975	<0.001	0.063	0.449
Dental	-138.041	22.943	184.607	-6.017	<0.001	0.063	0.449
Interdental	-67.012	39.649	180.436	-1.690	0.093	0.063	0.449
Postalveolar	8.127	22.231	182.542	0.366	0.715	0.063	0.449
Affricate /tʃ/							
(Intercept)	-925.884	16.546	4.937	-55.960	<0.001	0.047	0.268
Dental	-97.885	18.200	184.194	-5.378	<0.001	0.047	0.268
Interdental	-93.504	31.477	179.742	-2.971	0.003	0.047	0.268
Postalveolar	9.020	17.702	184.084	0.510	0.611	0.047	0.268
Fricative /z/							
(Intercept)	-1020.920	17.137	54.483	-59.573	<0.001	0.052	0.401
Dental	-177.085	33.227	187.814	-5.330	<0.001	0.052	0.401
Interdental	-29.150	56.692	178.406	-0.514	0.608	0.052	0.401
Postalveolar	15.197	31.784	180.463	0.478	0.633	0.052	0.401

FNE23

Integrated Fricative Noise Energy between FF2 (4 kHz) and FF3 (5.5 kHz).

Table S30. Linear Mixed-Effects model results for FNE23 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-1004.939	28.332	15.247	-35.470	<0.001	0.012	0.423
Dental	37.491	19.619	183.276	1.911	0.058	0.012	0.423
Interdental	-36.158	33.793	177.520	-1.070	0.286	0.012	0.423
Postalveolar	-40.638	18.982	180.747	-2.141	0.034	0.012	0.423
Affricate /tʃ/							
(Intercept)	-1051.930	25.288	4.009	-41.598	<0.001	0.011	0.289
Dental	45.767	21.617	183.405	2.117	0.036	0.011	0.289
Interdental	-44.985	37.419	179.637	-1.202	0.231	0.011	0.289
Postalveolar	-16.221	21.025	183.342	-0.771	0.441	0.011	0.289
Fricative /z/							
(Intercept)	-1169.569	19.734	37.550	-59.267	<0.001	0.007	0.314
Dental	52.212	34.998	187.478	1.492	0.137	0.007	0.314
Interdental	46.391	59.489	176.684	0.780	0.437	0.007	0.314
Postalveolar	-30.918	33.382	179.359	-0.926	0.356	0.007	0.314

FNE34

Integrated Fricative Noise Energy between FF3 (5.5 kHz) and FF4 (7.5 kHz).

Table S31. Linear Mixed-Effects model results for FNE34 (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	-1554.536	35.142	19.616	-44.236	<0.001	0.019	0.445
Dental	80.909	35.414	182.147	2.285	0.023	0.019	0.445
Interdental	48.498	61.190	177.931	0.793	0.429	0.019	0.445
Postalveolar	-66.164	34.313	180.075	-1.928	0.055	0.019	0.445
Affricate /tʃ/							
(Intercept)	-1610.767	30.657	5.339	-52.542	<0.001	0.020	0.356
Dental	86.990	35.690	183.991	2.437	0.016	0.020	0.356
Interdental	49.427	61.881	181.417	0.799	0.425	0.020	0.356
Postalveolar	-57.306	34.715	183.990	-1.651	0.100	0.020	0.356
Fricative /z/							
(Intercept)	-1821.219	29.185	31.814	-62.402	<0.001	0.022	0.320
Dental	134.549	50.200	187.796	2.680	0.008	0.022	0.320
Interdental	179.771	85.294	176.829	2.108	0.036	0.022	0.320
Postalveolar	-41.643	47.867	179.581	-0.870	0.385	0.022	0.320

FNER(12,23)

Ratio of FNE12 to FNE23 (spectral balance between the 2–4 kHz and 4–5.5 kHz integrated bands).

Table S32. Linear Mixed-Effects model results for FNER(12,23) (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	0.897	0.016	65.085	56.022	<0.001	0.101	0.516
Dental	0.180	0.028	183.203	6.364	<0.001	0.101	0.516
Interdental	0.020	0.049	180.121	0.404	0.687	0.101	0.516
Postalveolar	-0.047	0.027	181.571	-1.724	0.086	0.101	0.516
Affricate /tʃ/							
(Intercept)	0.899	0.016	8.033	56.417	<0.001	0.082	0.429
Dental	0.141	0.023	184.207	6.121	<0.001	0.082	0.429
Interdental	0.038	0.040	182.089	0.947	0.345	0.082	0.429
Postalveolar	-0.022	0.022	184.222	-0.991	0.323	0.082	0.429
Fricative /z/							
(Intercept)	0.892	0.018	27.373	50.857	<0.001	0.123	0.550
Dental	0.207	0.029	187.878	7.218	<0.001	0.123	0.550
Interdental	0.052	0.049	180.354	1.059	0.291	0.123	0.550
Postalveolar	-0.045	0.028	181.701	-1.646	0.101	0.123	0.550

FNER(12,34)

Ratio of FNE12 to FNE34 (spectral balance between the 2–4 kHz and 5.5–7.5 kHz integrated bands).

Table S33. Linear Mixed-Effects model results for FNER(12,34) (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	0.583	0.011	158.339	51.156	<0.001	0.102	0.565
Dental	0.139	0.024	182.940	5.793	<0.001	0.102	0.565
Interdental	0.081	0.041	180.452	1.957	0.052	0.102	0.565
Postalveolar	-0.031	0.023	181.578	-1.322	0.188	0.102	0.565
Affricate /tʃ/							
(Intercept)	0.589	0.009	55.707	65.011	<0.001	0.099	0.462
Dental	0.108	0.019	184.057	5.807	<0.001	0.099	0.462
Interdental	0.089	0.032	182.230	2.736	0.007	0.099	0.462
Postalveolar	-0.026	0.018	184.084	-1.451	0.148	0.099	0.462
Fricative /z/							
(Intercept)	0.571	0.014	34.565	40.590	<0.001	0.137	0.612
Dental	0.171	0.024	185.956	7.048	<0.001	0.137	0.612
Interdental	0.099	0.042	179.503	2.374	0.019	0.137	0.612
Postalveolar	-0.029	0.023	180.532	-1.227	0.222	0.137	0.612

FNER(23,34)

Ratio of FNE23 to FNE34 (spectral balance between the 4–5.5 kHz and 5.5–7.5 kHz integrated bands).

Table S34. Linear Mixed-Effects model results for FNER(23,34) (/s/, /tʃ/, /z/).

Term	Estimate	SE	df	t-val	p-val	Marg. R ²	Cond. R ²
Fricative /s/							
(Intercept)	0.654	0.008	101.803	87.125	<0.001	0.017	0.446
Dental	0.018	0.015	183.427	1.235	0.218	0.017	0.446
Interdental	0.059	0.025	180.180	2.315	0.022	0.017	0.446
Postalveolar	-0.001	0.014	181.723	-0.060	0.953	0.017	0.446
Affricate /tʃ/							
(Intercept)	0.659	0.009	7.958	74.967	<0.001	0.026	0.430
Dental	0.013	0.013	183.562	1.028	0.306	0.026	0.430
Interdental	0.058	0.022	181.745	2.603	0.010	0.026	0.430
Postalveolar	-0.014	0.012	183.583	-1.132	0.259	0.026	0.430
Fricative /z/							
(Intercept)	0.646	0.006	173.904	101.676	<0.001	0.023	0.445
Dental	0.029	0.014	184.627	2.049	0.042	0.023	0.445
Interdental	0.056	0.024	176.613	2.291	0.023	0.023	0.445
Postalveolar	0.000	0.014	178.163	-0.036	0.971	0.023	0.445